Paper or pixels?

What a semester of long-form reading tells us about comprehension on screens

Clara Lindqvist · Daniel Osei · Ines Moreau

In this talk

1

The question

Why ask about paper again, after two decades of studies.

2

The study

A semester, 212 students and the same eight essays in two media.

3

The findings

A gap that grows with length, and what readers did about it.

1

Part 1

The question

Twenty years of studies, one stubborn gap

- Meta-analyses agree: readers understand informational text a little better on paper (Delgado et al. 2018; Clinton 2019).
- The gap is small, but it has not closed as screens improved.
- Almost every study uses short passages read once, in a lab.

−0.21

pooled effect size for screen versus paper, informational texts under time pressure (Delgado et al. 2018)

“

On paper I know where I am in the argument. On a screen I only know where I am on the page.

Participant 27, second-year history student

Exit interview, December 2025

2

Part 2

The study

Reading where students actually read

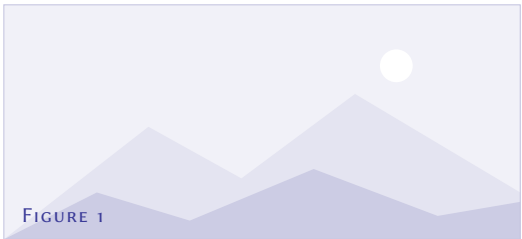


Figure 1

The reading room of the university library, where half of the sessions took place.

- 212 undergraduates, 14 weeks
- Eight essays of 2 000 to 9 000 words
- Each essay read on paper by one half of the class and on a tablet by the other
- Comprehension tested a week later, not the same day

Two conditions, one text

Paper

Printed reader

- Stapled A4 booklet, 11 pt serif
- Pencil margins allowed
- Readers keep the booklet for the week

Screen

Tablet reader

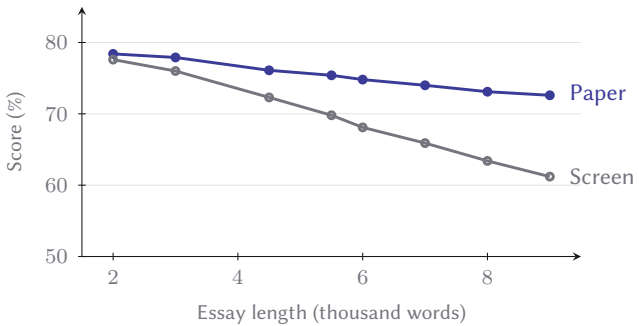
- The same layout as a PDF on a 10-inch tablet
- Highlighting and notes allowed
- Readers keep the file for the week

3

Part 3

The findings

The longer the essay, the wider the gap



Below 3 000 words the two media are indistinguishable. Past 6 000 words, screen readers lose about two points per thousand words.

Mean delayed-test score, $n = 212$.

What readers did with the text

	Paper	Screen	Difference
Reading time per essay (min)	41.2	36.8	−4.4
Annotations per 1 000 words	6.3	2.1	−4.2
Returns to an earlier page	9.7	3.4	−6.3
Delayed-test score (%)	75.1	69.6	−5.5
Confidence in own score (%)	71.0	78.3	+7.3

Screen readers read faster, marked less, and were more confident in a lower score: the calibration gap described by Singer and Alexander (2017).

What we take from it

1. The medium matters most for long, argued texts, the kind a degree is built on.
2. The cost comes from navigation: readers who go back keep their scores.
3. Overconfidence, not attention, is the risk worth teaching against.

Consistent with the slower, recursive reading that Wolf (2018) argues screens discourage, and with Mangen et al. (2013).

References

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Thank you.

Questions, data and materials

c.lindqvist@northfield.example · osf.example/paper-or-pixels